

**BRAC University**  
**Department of Computer Science and Engineering**

Examination: Quiz 02

Duration: 20 Minutes

Semester: Summer 26

Full Marks: 15

CSE420: Compiler Design

|                       |     |          |
|-----------------------|-----|----------|
| Name: <b>SOLUTION</b> | ID: | Section: |
|-----------------------|-----|----------|

1. Construct the LR(0) parse table for the grammar and automaton given below:

1.  $STMT \rightarrow CLASS\ ID\ \{ \ PRGRM\ \}$

2.  $STMT \rightarrow \epsilon$

3.  $PRGRM \rightarrow INT\ ID$

4.  $PRGRM \rightarrow \epsilon$

| State | Action  |    |    |    |         |        | Go-To |       |
|-------|---------|----|----|----|---------|--------|-------|-------|
|       | CLASS   | ID | {  | }  | INT     | \$     | STMT  | PRGRM |
| 0     | S1 / R2 | R2 | R2 | R2 | R2      | R2     | 8     |       |
| 1     |         | S2 |    |    |         |        |       |       |
| 2     |         |    | S3 |    |         |        |       |       |
| 3     | R4      | R4 | R4 | R4 | S6 / R4 | R4     |       | 4     |
| 4     |         |    |    | S5 |         |        |       |       |
| 5     | R1      | R1 | R1 | R1 | R1      | R1     |       |       |
| 6     |         | S7 |    |    |         |        |       |       |
| 7     | R3      | R3 | R3 | R3 | R3      | R3     |       |       |
| 8     |         |    |    |    |         | Accept |       |       |